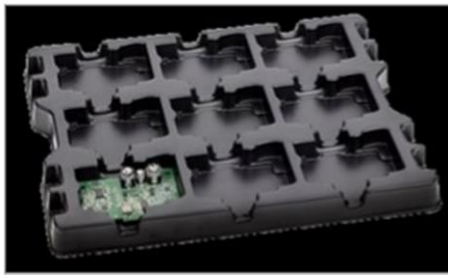
## 1. OVERVIEW

The following work instruction describes the process of handling ESD-sensitive parts. The workflow is to be followed by the TX5 material pickers.

## 2. GENERAL ESD HANDLING GUIDELINES

1. The ESD components can be identified by the following labels

2. Additionally, the ESD-sensitive components can be identified by the following packaging



- ✓ When the part looks like an ESD-sensitive part, it should be handled like one.
- ✓ Majority of ESD-sensitive components can be distinguished by an exposed PCB board.

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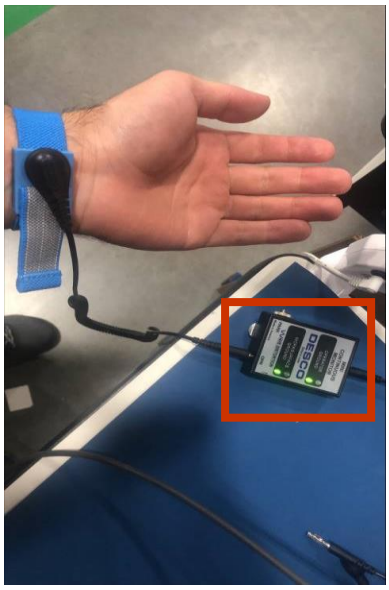
3. When picking the ESD components from the OEM packaging, the associate needs to bring the whole case to the ESD station and put the grounding wristband on prior to handling the units.



✓ The green light on the ESD monitoring device will indicate if the associate and workstation are grounded.

4. The unbagged components should never touch each other during handling.

5. The ESD components need to be packaged in static dissipative/protective packaging prior to leaving the ESD station.

6. Every station should be cleaned using a non-silicone cleaner.

## 3. BULK PACK ITEMS- JUMBLED IN BAG OR BOX FROM THE SUPPLIER

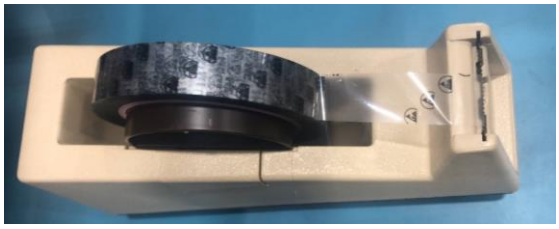
**1.** The bulk pack ESD items can be identified as parts packaged in the OEM box/bag without individual trays/dividers.

**2.** When picking a partial quantity (less than a full case), the associate can open the box to peek inside.

**3.** If the parts from OEM supplier have no individual unit packaging, they can be packed the same way in a single bag/box.

- ✓ It is best practice for the components to be nicely stacked in a box or neatly put in a bag.
- ✓ When packaging the parts in a mailer box, wrap them in an ESD-safe bubble wrap (Red Wrap).

**4.** Package the parts in a single mailer box/bag. If necessary, seal the bag with the ESD safe tape.



- ✓ Regular tape should never be used to tape the ESD-sensitive parts. Regular tape generates large amounts of static charge.

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**5.** Print out the part labels Refer to *Printing TX5 Bulk Parts Label* Attach the labels to the packaging.

✓ If parts are not shipped in supplier packaging with the labels available, then individual and overpack labels must be created and put on the outside of each packaging for serialized items, and an overpack only label must be created for non-serialized items.

## 4. PARTS PACKAGED IN TRAYS/LAYERS

**1.** The ESD items packaged in trays/layers can be identified as the items packaged in the OEM box, separated in trays or carton layers (multiple items or a single item per a tray/layer slot)

**2.** When picking a partial quantity (less than a full case), the associate can open the box to peek inside.

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**3.** Whenever possible, the parts should be picked and secured in a tray/layer.

**4.** If picking whole trays/layers is not possible (the pick quantity is less than the tray/layer quantity, or parts can't be secured in a tray/layer) and the parts don't have individual packaging from OEM, the associate needs to place the contents of each tray/layer slot in individual poly bags. Seal the poly bags with ESD-safe tape if necessary.

**5.** If a part is serialized, print, and attach the 2x2. Refer to Create and Print TX5 Parts Labels for each individual small poly bag.

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**6.** Place the individual bags in an overpack bag/box and seal the bag with ESD tape if necessary.

- ✓ Regular tape should never be used to tape the ESD-sensitive bags. Regular tape generates large amounts of static charge.

**7.** Print and attach an overpack label, *Refer to Printing TX5 Parts Labels.*

- ✓ If parts are not shipped in supplier packaging with the labels available, then individual and overpack labels must be created and put on the outside of each packaging for serialized items, and an overpack only label must be created for non-serialized items.

## 5. ESD SENSITIVE SCAN ENGINES

**1.** The ESD-sensitive scan engines can be identified as large ESD parts, with the notation "Engine" in the SKU description.

**2.** When picking a partial quantity (less than a full case), the associate can open the box to peek inside.

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**3.** Whenever possible, parts should be picked and secured in a tray.

**4.** If picking by whole trays is not possible (the pick quantity is less than the tray quantity, or parts can't be secured in a tray) and the parts don't have individual packaging from OEM, the associate needs to place each scan engine in individual static shielding bags. Seal the static shielding bags with ESD-safe tape if necessary.

✓ When handling the scan engines, the associate must wear ESD finger cots.

**5.** Print and attach the 2x2 labels Printing *TX5 Bulk Parts Labels* to each individual static shielding bag.

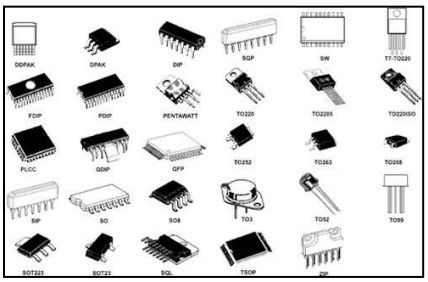
**6.** Individual bags must be wrapped in an ESD bubble wrap and placed in an appropriately sized mailer box. Multiple scan engines can be placed in a single mailer box, but they must be individually wrapped in a bubble wrap.

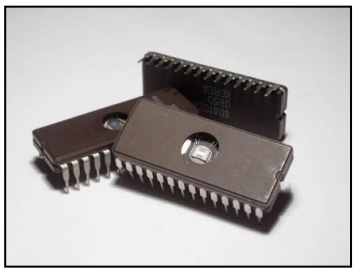
**7.** Print and attach an overpack label, Refer to *Create and Print TX5 Bulk Parts Label*.

- ✓ If parts are not shipped in supplier packaging with the labels available, then individual and overpack labels must be created and put on the outside of each packaging for serialized items, and an overpack only label must be created for non-serialized items.

## 6. ESD SENSITIVE COMPONENT LEVEL PARTS

**1.** The ESD sensitive component level parts can be identified as ESD parts that are installed on the PCB boards (BGAs, ICs, resistors, capacitors, inductors)





**2.** When picking a partial quantity (less than a full case), the associate can open the box to peek inside.

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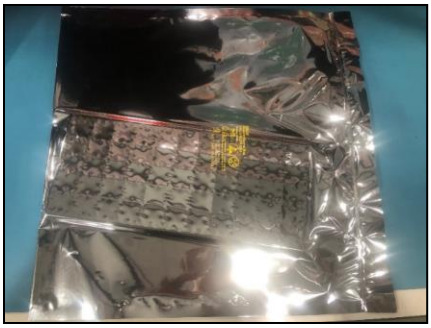
**3.** Whenever possible, parts should be picked and secured in a tray/tube/strip.

**4.** If picking by whole trays/tubes/strips is not possible (the pick quantity is less than the inner-pack quantity, or parts can't be secured in an inner-pack) and the parts don't have individual packaging from OEM, the associate will need to do one of the following:

A) Place a BGA/IC component on a tray in a moisture-resistant static shielding bag together with the moisture strip and a desiccant. Seal the bag with the vacuum sealer refer to Vacuum Sealing Operation.



B) Place each component (not BGA or IC) in individual static shielding bags. Seal the static shielding bags with ESD safe tape if necessary.

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**5.** Print and attach the 2x2 labels, refer to *Create and Print TX5 Bulk Parts Label* to each individual static shielding bag.

**6.** Individual bags (not BGA or IC) must be wrapped in an ESD bubble wrap and placed in an appropriately sized mailer box. Multiple components can be placed in a single mailer box, but they must be individually wrapped in a bubble wrap.

**7.** Print and attach an overpack label, refer to *Printing TX5 Bulk Parts Labels*.

✓ Note: If parts are not shipped in supplier packaging with the labels available, then individual and overpack labels must be created and put on the outside of each packaging for serialized items, and an overpack-only label must be created for non-serialized items.

## 7. REFERENCES

| Document # | Document Title |
|---|---|
| | Create and Print TX5 Bulk Parts Label Rev01.pdf |
| | Vacuum Sealing Operation Rev02.pdf |

## 8. REVISION HISTORY

The following table lists all revisions (including the original document) to this procedure, the date, and the reason for the revision.

| Rev. | Rev. Date | Description of Change | Revised By |
|---|---|---|---|
| 01 | 08/24/22 | Original issue | P. Eroshenko |
| 02 | 03/07/23 | Rebranded | A.Cabrera |

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